

Practical 1: Develop python program to import and export data using Numpy, Pandas library function.

1. NumPy — Import and Export Data

```
import numpy as np
# ----- EXPORT (Save) -----
# Create sample data
data = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])
np.save('data.npy', data)
# Save as .csv (text format - human readable)
np.savetxt('data.csv', data, delimiter=',', fmt='%d')
# Save as .txt
np.savetxt('data.txt', data, delimiter=' ', fmt='%d')
print("Data exported successfully!")
print(data)
# ----- IMPORT (Load) -----
# Load .npy file
loaded_npy = np.load('data.npy')
print("\nLoaded from .npy:\n", loaded_npy)
# Load .csv file
loaded_csv = np.loadtxt('data.csv', delimiter=',')
print("\nLoaded from .csv:\n", loaded_csv)
# Load .txt file
loaded_txt = np.loadtxt('data.txt', delimiter=' ')
print("\nLoaded from .txt:\n", loaded_txt)
```

Output:

```
D:\n_nagpure\venv\Scripts\pyt
Data exported successfully!
[[1 2 3]
 [4 5 6]
 [7 8 9]]

Loaded from .npy:
[[1 2 3]
 [4 5 6]
 [7 8 9]]

Loaded from .csv:
[[1. 2. 3.]
 [4. 5. 6.]
 [7. 8. 9.]]

Loaded from .txt:
[[1. 2. 3.]
 [4. 5. 6.]
 [7. 8. 9.]]
```

2. Pandas — Import and Export Data

```
import pandas as pd
# ----- Create Sample DataFrame -----
df = pd.DataFrame({
    'RollNo': [1, 2, 3, 4],
    'Name': ['Aditi', 'Rahul', 'Sneha', 'Karan'],
    'Marks': [85, 78, 92, 67]
})
# ----- EXPORT (Save) -----
df.to_csv('students.csv', index=False)    # CSV
df.to_excel('students.xlsx', index=False) # Excel (needs openpyxl)
df.to_json('students.json', orient='records') # JSON
print("DataFrame exported successfully!")
print(df)
# ----- IMPORT (Load) -----
df_csv = pd.read_csv('students.csv')
print("\nLoaded from CSV:\n", df_csv)
df_excel = pd.read_excel('students.xlsx')
print("\nLoaded from Excel:\n", df_excel)
df_json = pd.read_json('students.json')
print("\nLoaded from JSON:\n", df_json)
```

Output:

DataFrame exported successfully!

	RollNo	Name	Marks
0	1	Aditi	85
1	2	Rahul	78
2	3	Sneha	92
3	4	Karan	67

Loaded from CSV:

	RollNo	Name	Marks
0	1	Aditi	85
1	2	Rahul	78
2	3	Sneha	92
3	4	Karan	67

Loaded from Excel:

	RollNo	Name	Marks
0	1	Aditi	85
1	2	Rahul	78
2	3	Sneha	92
3	4	Karan	67

Loaded from JSON:

	RollNo	Name	Marks
0	1	Aditi	85
1	2	Rahul	78
2	3	Sneha	92
3	4	Karan	67